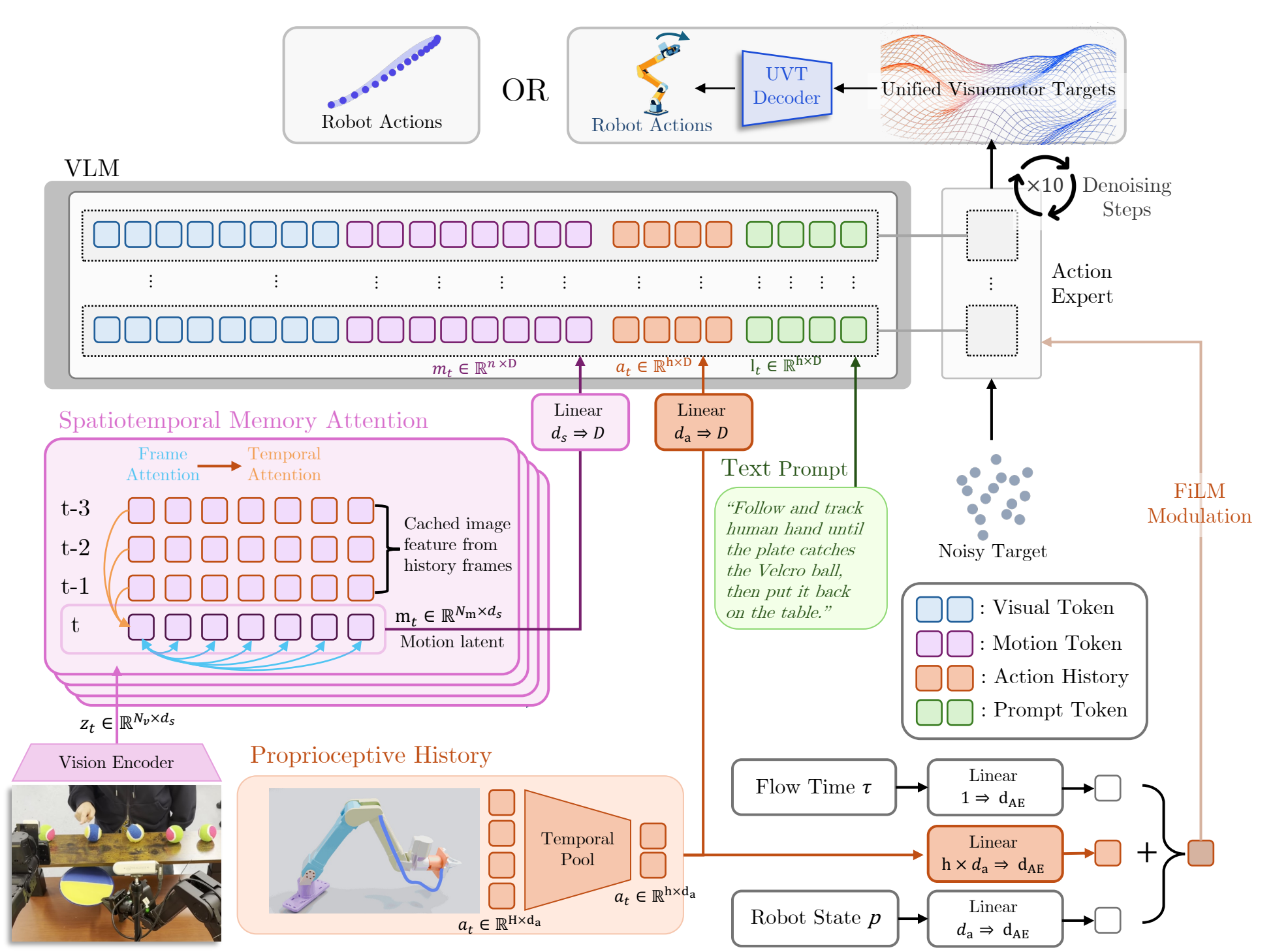

Robot Actions
OR
Robot Actions
UVT Decoder
Unified Visuomotor Targets
VLM
×10
Denoising Steps
Action Expert
$m_t \in \mathbb{R}^{n \times D}$
$a_t \in \mathbb{R}^{h \times D}$
$l_t \in \mathbb{R}^{h \times D}$
Linear $d_s \Rightarrow D$
Linear $d_a \Rightarrow D$
Spatiotemporal Memory Attention
Frame Attention
Temporal Attention
t-3
t-2
t-1
t
Cached image feature from history frames
$m_t \in \mathbb{R}^{N_m \times d_s}$
Motion latent
$z_t \in \mathbb{R}^{N_v \times d_s}$
Vision Encoder
Text Prompt
"Follow and track human hand until the plate catches the Velcro ball, then put it back on the table."
Noisy Target
FiLM Modulation
: Visual Token
: Motion Token
: Action History
: Prompt Token
Proprioceptive History
Temporal Pool
$a_t \in \mathbb{R}^{H \times d_a}$
$a_t \in \mathbb{R}^{h \times d_a}$
Flow Time τ
Linear $1 \Rightarrow d_{AE}$
Linear $h \times d_a \Rightarrow d_{AE}$
Robot State p
Linear $d_a \Rightarrow d_{AE}$